50 Startups Mlr Notebook

Multiple Linear Regression — Startup Profit Prediction

This notebook builds a **Multiple Linear Regression** model to predict startup profitability based on spending in R&D, Administration, and Marketing.

I followed a structured, modular EDA pipeline:

Phase	Description
1	Import Libraries
2	Load Dataset
3	Shape & Structure
4	Data Types
5	Missing Values
6	Duplicate Check
7	Descriptive Statistics
8	Target Distribution
9	Feature Distributions
10	Scatter Plots (Features vs Target)
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Phase	Description
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23	Conclusion

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2 · Load Dataset

Read the CSV file into a DataFrame.

```
df = pd.read_csv('50_startups_dataset.csv')
print(f'Dataset loaded — {df.shape[0]} rows, {df.shape[1]} columns.')
```

Output:

```
Dataset loaded — 50 rows, 5 columns.
```

> ***AI Note:*** *Displays the first few rows of the 'df' DataFrame to provide a glimpse into the data.*

```
df.head()
```

Result:

```
   R&D Spend  Administration  Marketing Spend       State     Profit
0  165349.20       136897.80        471784.10    Pakistan  192261.83
1  162597.70       151377.59        443898.53  California  191792.06
2  153441.51       101145.55        407934.54    Istanbul  191050.39
3  144372.41       118671.85        383199.62    Pakistan  182901.99
4  142107.34        91391.77        366168.42    Istanbul  166187.94
```

> ***AI Note:*** *Displays the last few rows of the 'df' DataFrame for data inspection.*

```
df.tail()
```

Result:

```
    R&D Spend  Administration  Marketing Spend       State    Profit
45    1000.23       124153.04          1903.93    Pakistan  64926.08
46    1315.46       115816.21        297114.46    Istanbul  49490.75
47       0.00       135426.92             0.00  California  42559.73
48     542.05        51743.15             0.00    Pakistan  35673.41
49       0.00       116983.80         45173.06  California  14681.40
```

---

4 · Data Types

Verify which columns are numeric vs categorical.

> ***AI Note:*** *Displays the data types of each column in the 'df' DataFrame.*

```
df.dtypes
```

Result:

```
R&D Spend          float64
Administration     float64
Marketing Spend    float64
State                  str
Profit             float64
dtype: object
```

> ***AI Note:*** *Displays information about the 'df' DataFrame, including data types and memory usage.*

```
df.info()
```

Output:

```
<class 'pandas.DataFrame'>
RangeIndex: 50 entries, 0 to 49
Data columns (total 5 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   R&D Spend        50 non-null     float64
 1   Administration   50 non-null     float64
 2   Marketing Spend  50 non-null     float64
 3   State            50 non-null     str
 4   Profit           50 non-null     float64
dtypes: float64(4), str(1)
```

```
memory usage: 2.1 KB
```

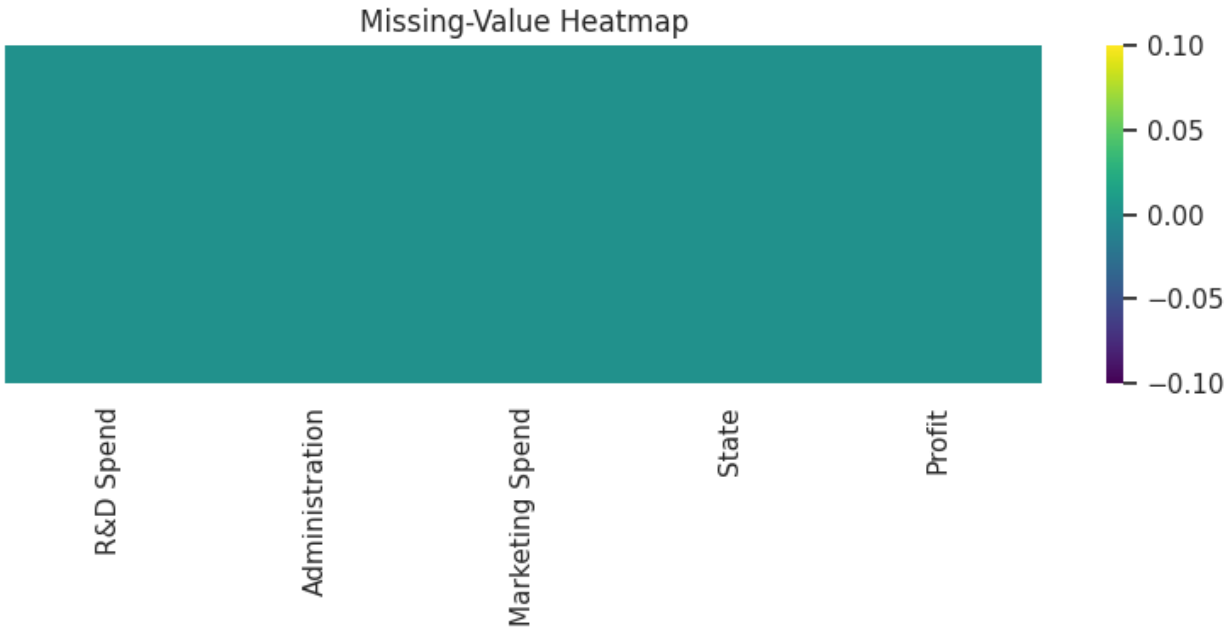
---

5 · Missing Values

Check for nulls in every column.

> ***Summary:*** *This cell calculates and displays the number of missing values in each column of the 'df' DataFrame, as well as the percentage of missing values in each column.*
>
> ***AI Note:*** *This cell generates a heatmap to visualize missing values in the 'df' DataFrame, providing a visual representation of data quality.*

```
plt.figure(figsize=(8, 4))
sns.heatmap(df.isnull(), cbar=True, yticklabels=False, cmap='viridis')
plt.title('Missing-Value Heatmap')
plt.tight_layout()
plt.show()
```



6 · Duplicate Check

Identify and count exact duplicate rows.

> ***AI Note:*** *This cell checks for duplicate rows in the 'df' DataFrame and displays the count. If duplicates are found, it also displays the duplicate rows.*

```
dup_count = df.duplicated().sum()
print(f'Duplicate rows: {dup_count}')

if dup_count > 0:
    print('\nDuplicate rows:')
    display(df[df.duplicated(keep=False)])
```

Output:

```
Duplicate rows: 0
```

> ***AI Note:*** *This cell provides a section title and a subheading for the descriptive statistics section, setting the context for the following analysis.*

---

7 · Descriptive Statistics

7.1 — Numerical Summary

> ***AI Note:*** *This cell displays descriptive statistics of the 'df' DataFrame, including count, mean, standard deviation, minimum, 25th percentile, median, 75th percentile, and maximum for each column.*

```
df.describe()
```

Result:

```
            R&D Spend  Administration  Marketing Spend         Profit
count       50.000000       50.000000        50.000000      50.000000
mean     73721.615600   121344.639600    211025.097800  112012.639200
std      45902.256482    28017.802755    122290.310726   40306.180338
min          0.000000    51283.140000         0.000000   14681.400000
25%      39936.370000   103730.875000    129300.132500   90138.902500
50%      73051.080000   122699.795000    212716.240000  107978.190000
```

```
75%    101602.800000   144842.180000    299469.085000  139765.977500
max    165349.200000   182645.560000    471784.100000  192261.830000
```

> ***AI Note:*** *Section title and subheading for the descriptive statistics section, setting the context for the following analysis.*

7.2 — Categorical Summary

> ***AI Note:*** *This cell displays the count, unique values, top value, and frequency of the 'State' column in the 'df' DataFrame.*

```
df.describe(include='object')
```

Output:

```
/tmp/ipykernel_70051/87514550.py:1: Pandas4Warning: For backward compatibility,
See https://pandas.pydata.org/docs/user_guide/migration-3-strings.html#string-mi
  df.describe(include='object')
```

Result:

```
           State
count         50
unique         3
top     Pakistan
freq          17
```

> ***AI Note:*** *This cell provides a section title and subheading for the target distribution section, setting the context for the following analysis.*

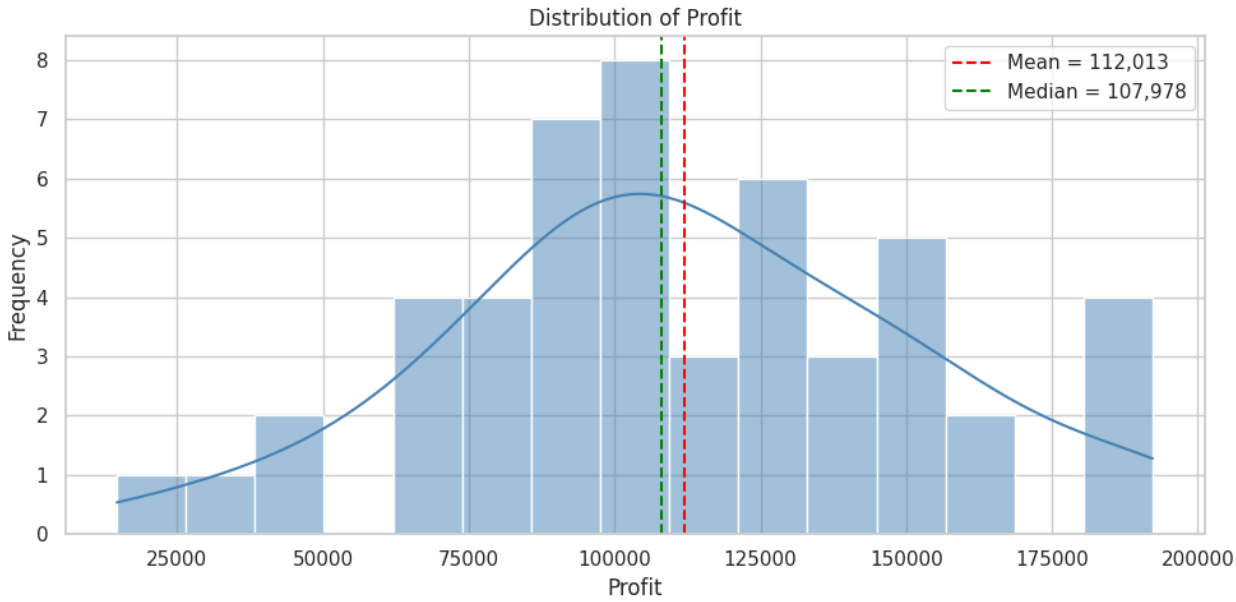
---

8 · Target Distribution

Visualize the distribution of the target variable — **Profit**.

> ***AI Note:*** *This cell generates a histogram of the 'Profit' column in the 'df' DataFrame, including a kernel density estimate (KDE) and vertical lines representing the mean and median values.*

```
plt.figure(figsize=(10, 5))
sns.histplot(df['Profit'], kde=True, bins=15, color='steelblue')
plt.axvline(df['Profit'].mean(), color='red', linestyle='--', label=f"Mean = {df
plt.axvline(df['Profit'].median(), color='green', linestyle='--', label=f"Median
plt.title('Distribution of Profit')
plt.xlabel('Profit')
plt.ylabel('Frequency')
plt.legend()
plt.tight_layout()
plt.show()
```



> ***AI Note:*** *Provides a section title and subheading for the feature distributions section, setting the context for the following analysis.*

---

9 · Feature Distributions

9.1 — Histograms of all numeric features

> ***AI Note:*** *This cell generates histograms for each numeric column in the 'df' DataFrame, providing a visual representation of their distributions.*

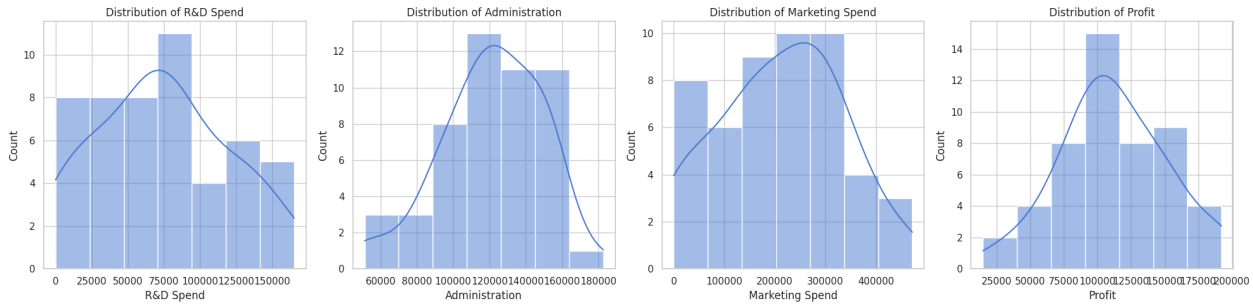
```
numeric_cols = df.select_dtypes(include='number').columns.tolist()

fig, axes = plt.subplots(1, len(numeric_cols), figsize=(5 * len(numeric_cols), 5

for ax, col in zip(axes, numeric_cols):
```

```
    sns.histplot(df[col], kde=True, ax=ax)
    ax.set_title(f'Distribution of {col}')

plt.tight_layout()
plt.show()
```



> ***AI Note:*** *Section title and subheading for the Box Plots section, setting the context for the following analysis on outlier detection.*

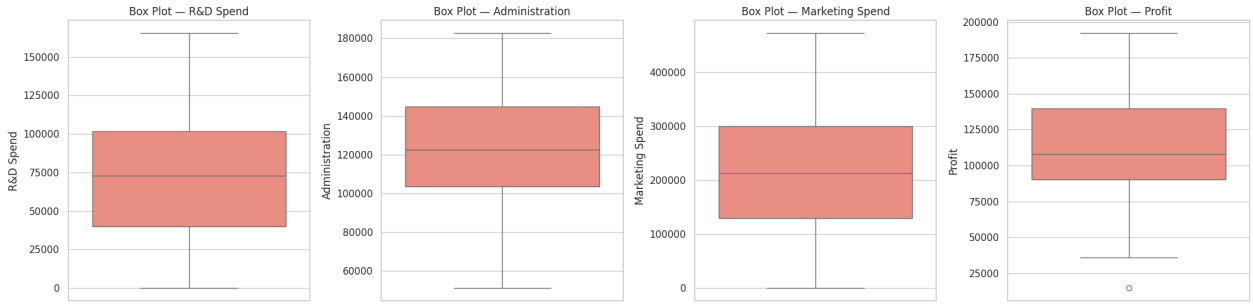
9.2 — Box Plots (outlier detection)

> ***AI Note:*** *This cell generates a box plot for each numeric column in the 'df' DataFrame, providing a visual representation of the distribution of each feature and helping to identify potential outliers.*

```
fig, axes = plt.subplots(1, len(numeric_cols), figsize=(5 * len(numeric_cols), 5

for ax, col in zip(axes, numeric_cols):
    sns.boxplot(y=df[col], ax=ax, color='salmon')
    ax.set_title(f'Box Plot — {col}')

plt.tight_layout()
plt.show()
```



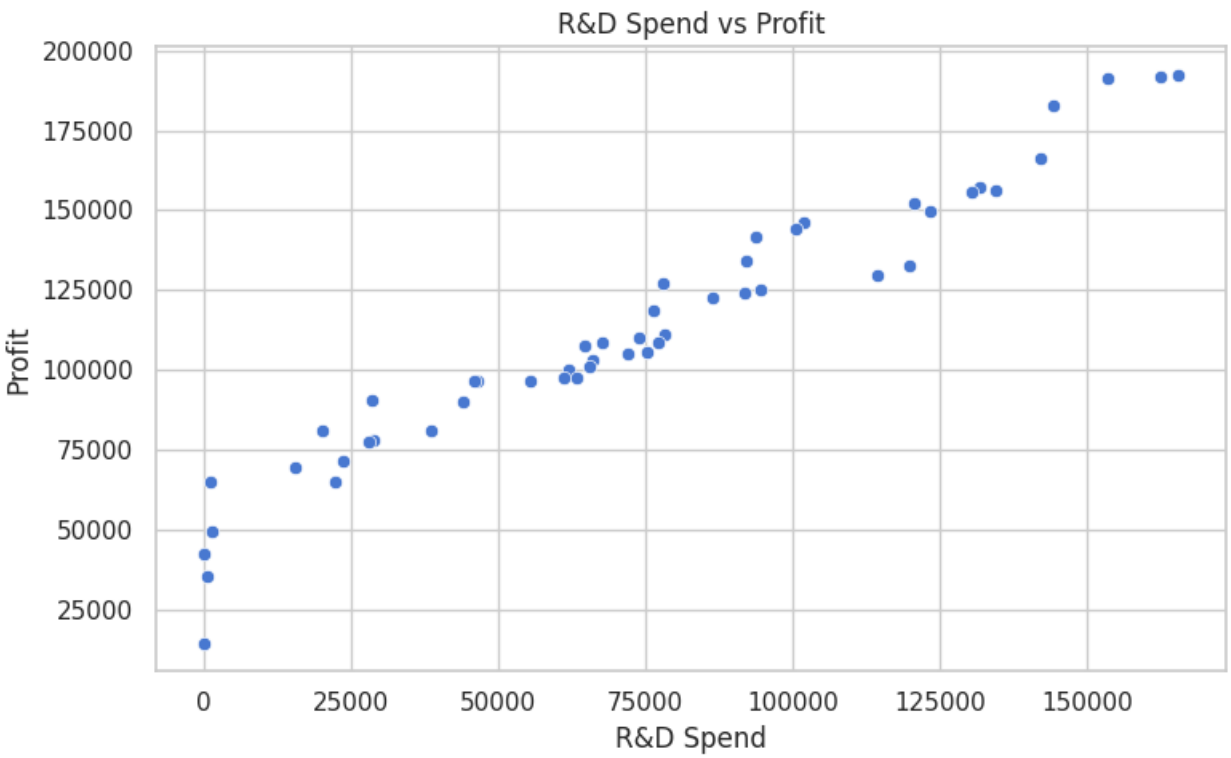
10 · Scatter Plots — Features vs Target

Examine the linear relationship between each numeric feature and Profit.

10.1 — R&D Spend vs Profit

> ***AI Note:*** *This cell generates a scatter plot to visualize the relationship between R&D Spend and Profit, providing insights into the potential impact of R&D investment on business outcomes.*

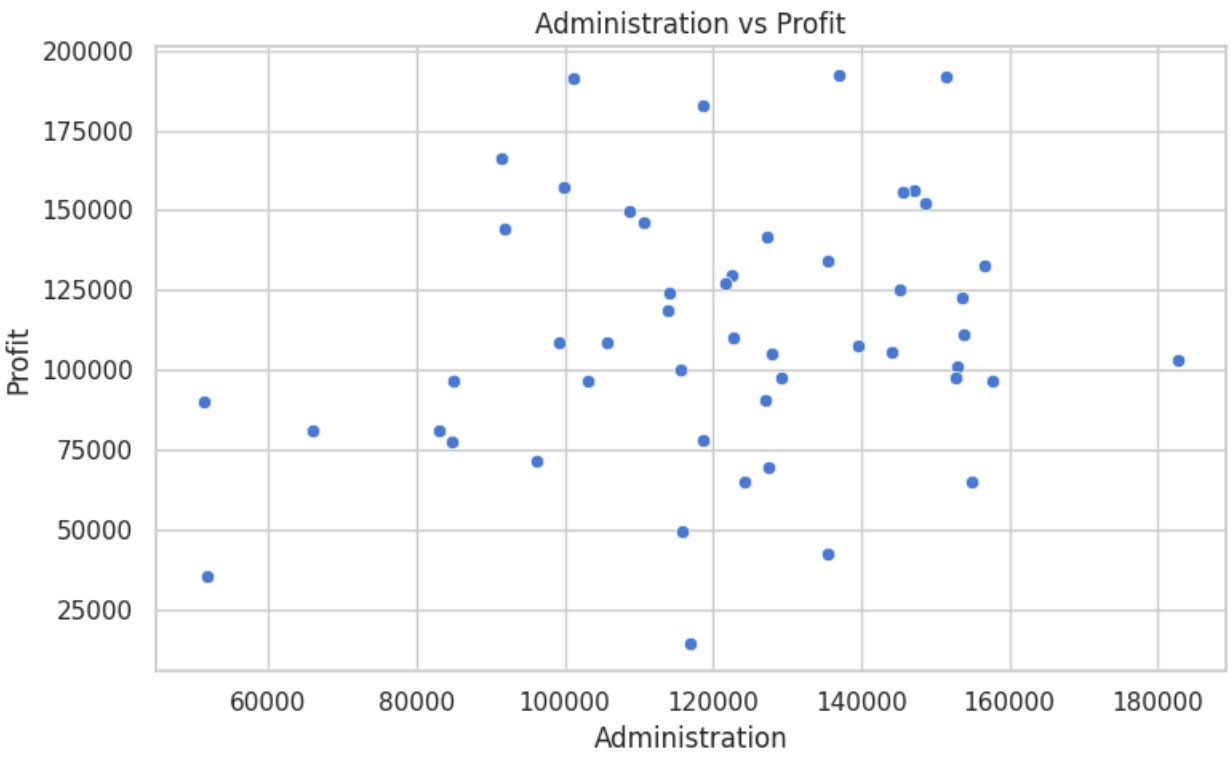
```
plt.figure(figsize=(8, 5))
sns.scatterplot(x='R&D Spend', y='Profit', data=df)
plt.title('R&D Spend vs Profit')
plt.tight_layout()
plt.show()
```



10.2 — Administration vs Profit

> ***AI Note:*** *This cell generates a scatter plot to visualize the relationship between Administration and Profit, providing insights into the potential impact of administrative costs on business outcomes.*

```
plt.figure(figsize=(8, 5))
sns.scatterplot(x='Administration', y='Profit', data=df)
plt.title('Administration vs Profit')
plt.tight_layout()
plt.show()
```



10.3 — Marketing Spend vs Profit

> ***AI Note:*** *This cell generates a scatter plot to visualize the relationship between Marketing Spend and Profit, providing insights into the potential impact of marketing investment on business outcomes.*

```
plt.figure(figsize=(8, 5))
sns.scatterplot(x='Marketing Spend', y='Profit', data=df)
plt.title('Marketing Spend vs Profit')
plt.tight_layout()
plt.show()
```

---

11 · Categorical Analysis

11.1 — State Value Counts

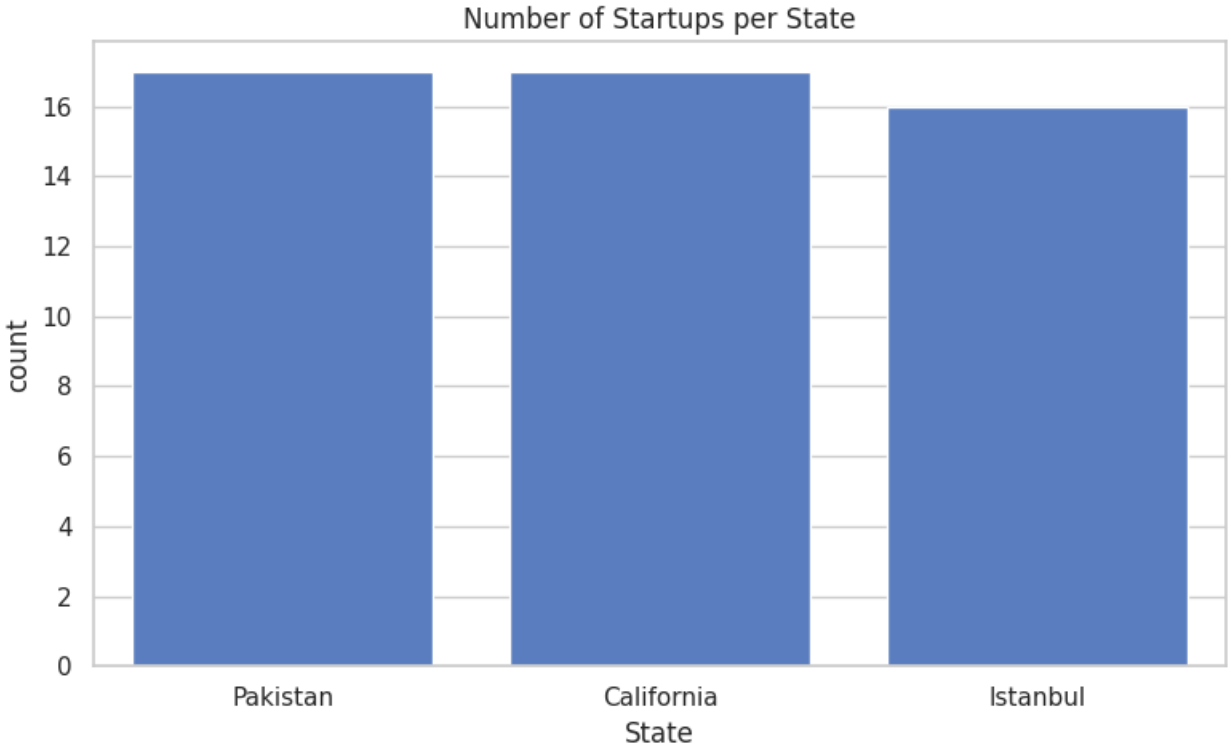
```
df['State'].value_counts()
```

Result:

```
State
Pakistan      17
California    17
Istanbul      16
Name: count, dtype: int64
```

> ***AI Note:*** *This cell generates a count plot to visualize the distribution of startups across different states, providing insights into regional business activity.*

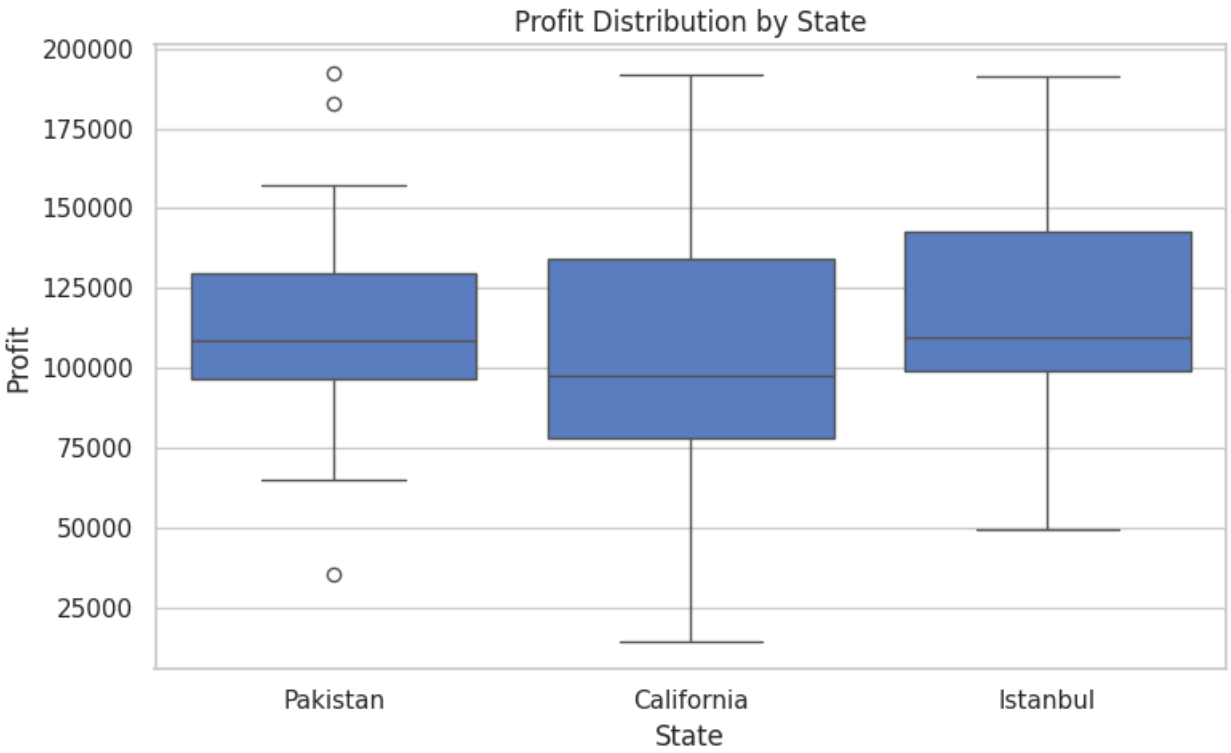
```
plt.figure(figsize=(8, 5))
sns.countplot(x='State', data=df, order=df['State'].value_counts().index)
plt.title('Number of Startups per State')
plt.tight_layout()
plt.show()
```



11.3 — Profit by State (Box Plot)

> ***AI Note:*** *Generates a box plot to visualize the distribution of profit across different states, helping to identify potential outliers and patterns in the data.*

```
plt.figure(figsize=(8, 5))
sns.boxplot(x='State', y='Profit', data=df)
plt.title('Profit Distribution by State')
plt.tight_layout()
plt.show()
```



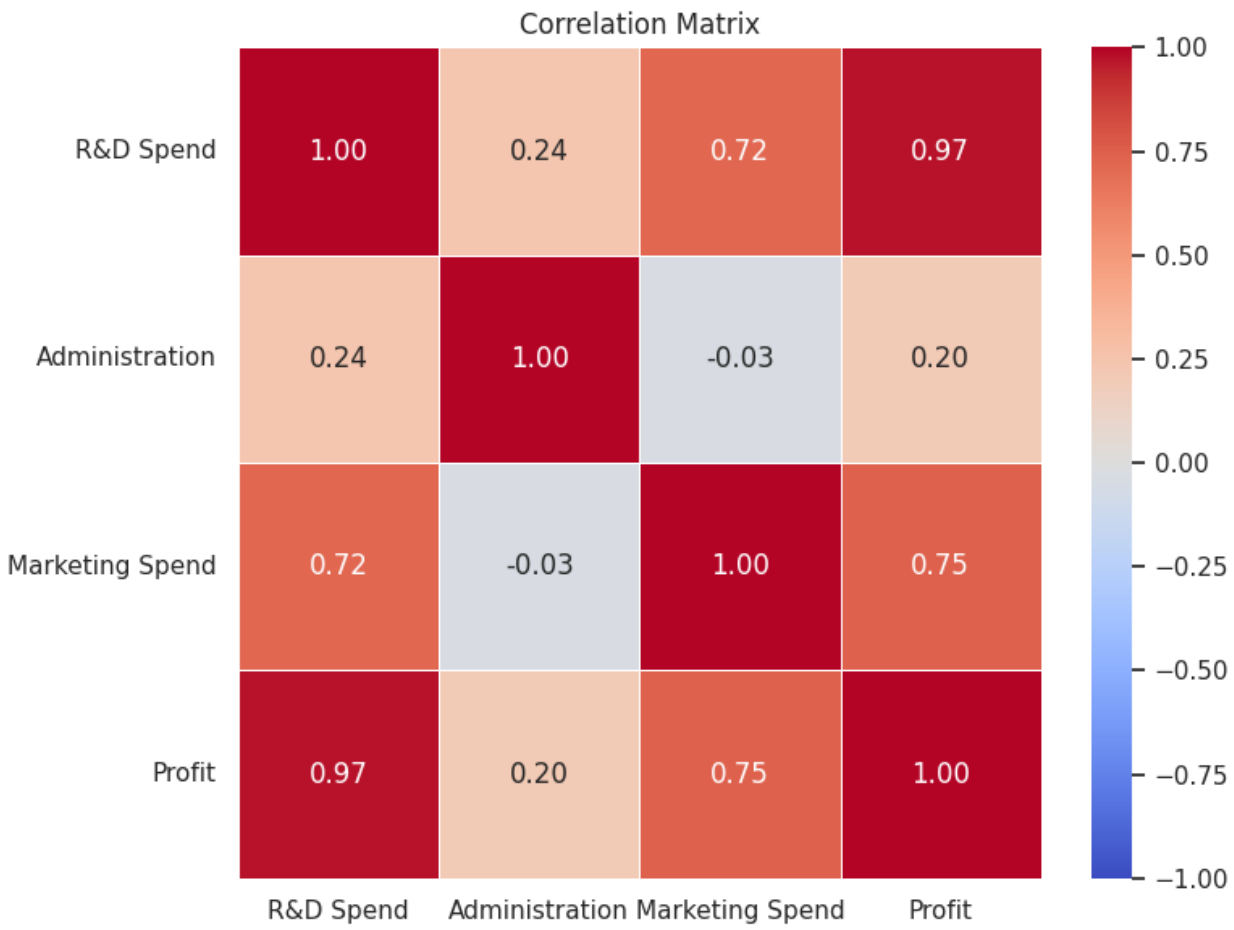
12 · Correlation Matrix

Examine pairwise correlations among numeric features.

> ***AI Note:*** *This cell generates a correlation matrix heatmap to visualize the relationships between numeric features in the 'df' DataFrame.*

```
corr = df.corr(numeric_only=True)

plt.figure(figsize=(8, 6))
sns.heatmap(corr, annot=True, cmap='coolwarm', fmt='.2f', linewidths=0.5, vmin=-
plt.title('Correlation Matrix')
plt.tight_layout()
plt.show()
```



> ***AI Note:*** *Section title and subheading for the correlation with target analysis, setting the context for the following analysis.*

12.1 — Correlation with Target

> ***AI Note:*** *Calculates the correlation between the 'Profit' column and other numeric features in the DataFrame, sorting the results in descending order of correlation.*

```
corr['Profit'].drop('Profit').sort_values(ascending=False)
```

Result:

```
R&D Spend          0.972900
Marketing Spend    0.747766
Administration     0.200717
Name: Profit, dtype: float64
```

> ***AI Note:*** *Sets the context for the following analysis on pairplot, providing a visual representation of the relationship between numeric variables and State.*

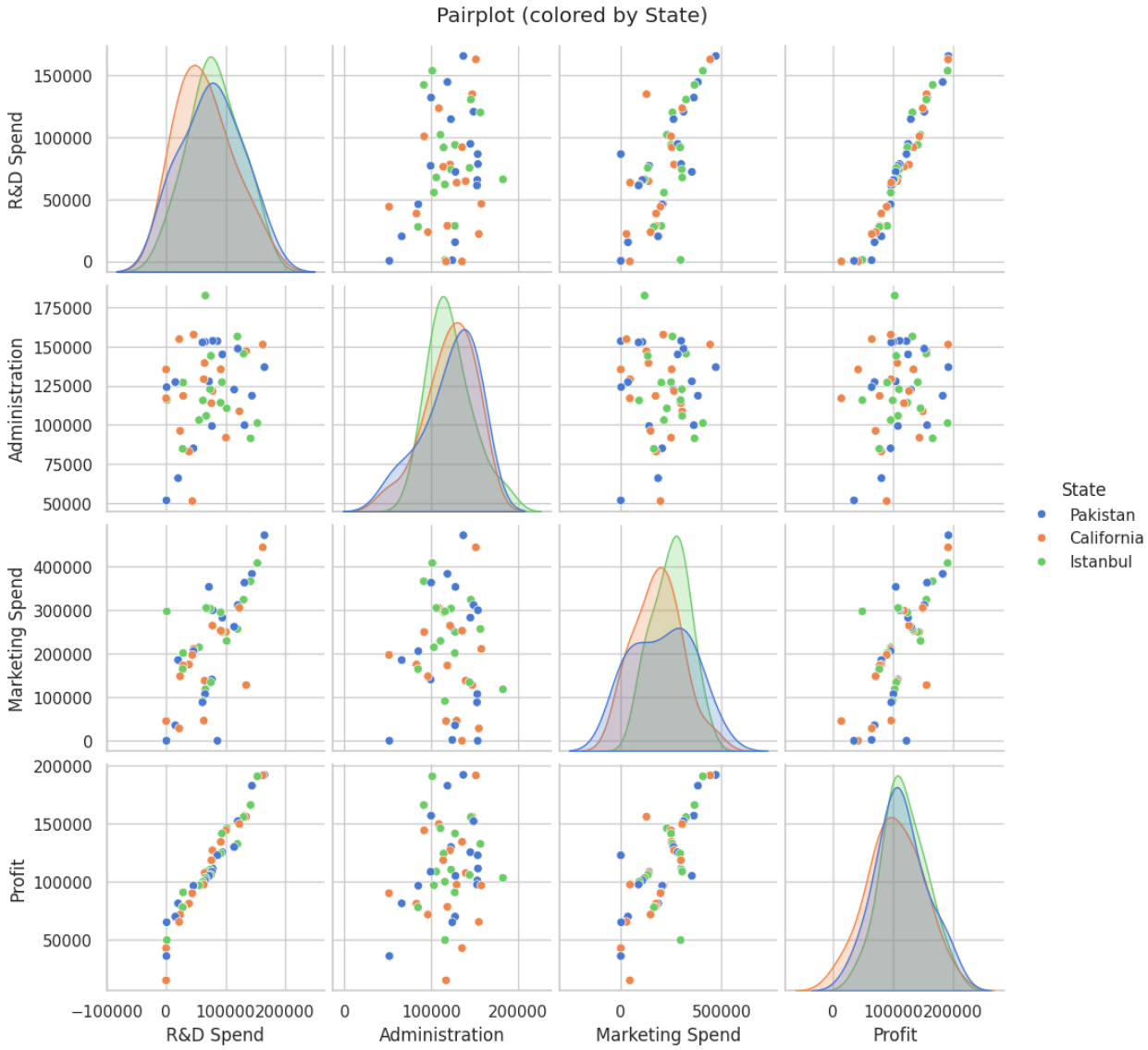
---

13 · Pairplot

Scatter matrix of all numeric variables, colored by State.

> ***AI Note:*** *Generates a pairplot to visualize the relationships between numeric features in the 'df' DataFrame, colored by the 'State' column, helping to identify patterns and correlations in the data.*

```
sns.pairplot(df, hue='State', diag_kind='kde')
plt.suptitle('Pairplot (colored by State)', y=1.02)
plt.show()
```



14 · One-Hot Encoding

Convert the categorical **State** column into dummy variables.

> ***AI Note:*** *This cell encodes the 'State' column in the DataFrame using one-hot encoding, dropping the first category to avoid multicollinearity.*

```
df_encoded = pd.get_dummies(df, columns=['State'], drop_first=True)
print(f'Columns after encoding: {df_encoded.columns.tolist()}')
df_encoded.head()
```

Output:

```
Columns after encoding: ['R&D Spend', 'Administration', 'Marketing Spend', 'Prof
```

Result:

```
   R&D Spend  Administration  Marketing Spend     Profit  State_Istanbul  \
0  165349.20       136897.80        471784.10  192261.83           False
1  162597.70       151377.59        443898.53  191792.06           False
2  153441.51       101145.55        407934.54  191050.39            True
3  144372.41       118671.85        383199.62  182901.99           False
4  142107.34        91391.77        366168.42  166187.94            True

   State_Pakistan
0            True
1           False
2           False
3            True
4           False
```

15 · Feature–Target Split

Separate independent variables (**X**) from the target variable (**y**).

> ***AI Note:*** *Provides context for the train-test split with a fixed random state for reproducibility.*

---

16 · Train–Test Split

80 / 20 split with a fixed random state for reproducibility.

> ***AI Note:*** *This cell performs a train-test split on the dataset using a fixed test size and random state for reproducibility.*

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

print(f'X_train: {X_train.shape}  |  X_test: {X_test.shape}')
print(f'y_train: {y_train.shape}  |  y_test: {y_test.shape}')
```

Output:

```
X_train: (40, 5)  |  X_test: (10, 5)
y_train: (40,)  |  y_test: (10,)
```

---

17 · Model Training

Fit a **Multiple Linear Regression** model on the training set.

> ***AI Note:*** *Trains a linear regression model on the training data and prints the intercept value.*

```
model = LinearRegression()
model.fit(X_train, y_train)

print('Model trained successfully.')
print(f'Intercept: {model.intercept_:,.2f}')
```

Output:

```
Model trained successfully.
Intercept: 54,028.04
```

> ***AI Note:*** *Provides context for the predictions analysis, setting the stage for the comparison of predicted and actual outcomes.*

---

18 · Predictions

Generate predictions on the test set and compare with actuals.

> ***AI Note:*** *This cell evaluates the performance of a model by comparing its predictions with actual values, providing insights into the model's accuracy.*

```
y_pred = model.predict(X_test)

comparison = pd.DataFrame({'Actual': y_test.values, 'Predicted': y_pred})
comparison['Error'] = comparison['Actual'] - comparison['Predicted']
comparison
```

Result:

```
      Actual      Predicted         Error
0  134307.35  126362.879083   7944.470917
1   81005.76   84608.453836  -3602.693836
2   99937.59   99677.494251    260.095749
3   64926.08   46357.460686  18568.619314
4  125370.37  128750.482885  -3380.112885
5   35673.41   50912.417419 -15239.007419
6  105733.54  109741.350327  -4007.810327
7  107404.34  100643.242816   6761.097184
8   97427.84   97599.275746   -171.435746
9  122776.86  113097.425244   9679.434756
```

---

19 · Evaluation Metrics

Quantify model performance using standard regression metrics.

> ***AI Note:*** *Evaluates the performance of a regression model by calculating R^2 score, mean absolute error, mean squared error, and root mean squared error.*

```
r2   = r2_score(y_test, y_pred)
mae  = mean_absolute_error(y_test, y_pred)
mse  = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)

print(f'R² Score                     : {r2:.4f}')
print(f'Mean Absolute Error  (MAE)  : {mae:,.2f}')
print(f'Mean Squared Error   (MSE)  : {mse:,.2f}')
print(f'Root Mean Squared Error (RMSE): {rmse:,.2f}')
```

Output:

```
R² Score                     : 0.8987
Mean Absolute Error  (MAE)  : 6,961.48
Mean Squared Error   (MSE)  : 82,010,363.04
Root Mean Squared Error (RMSE): 9,055.96
```

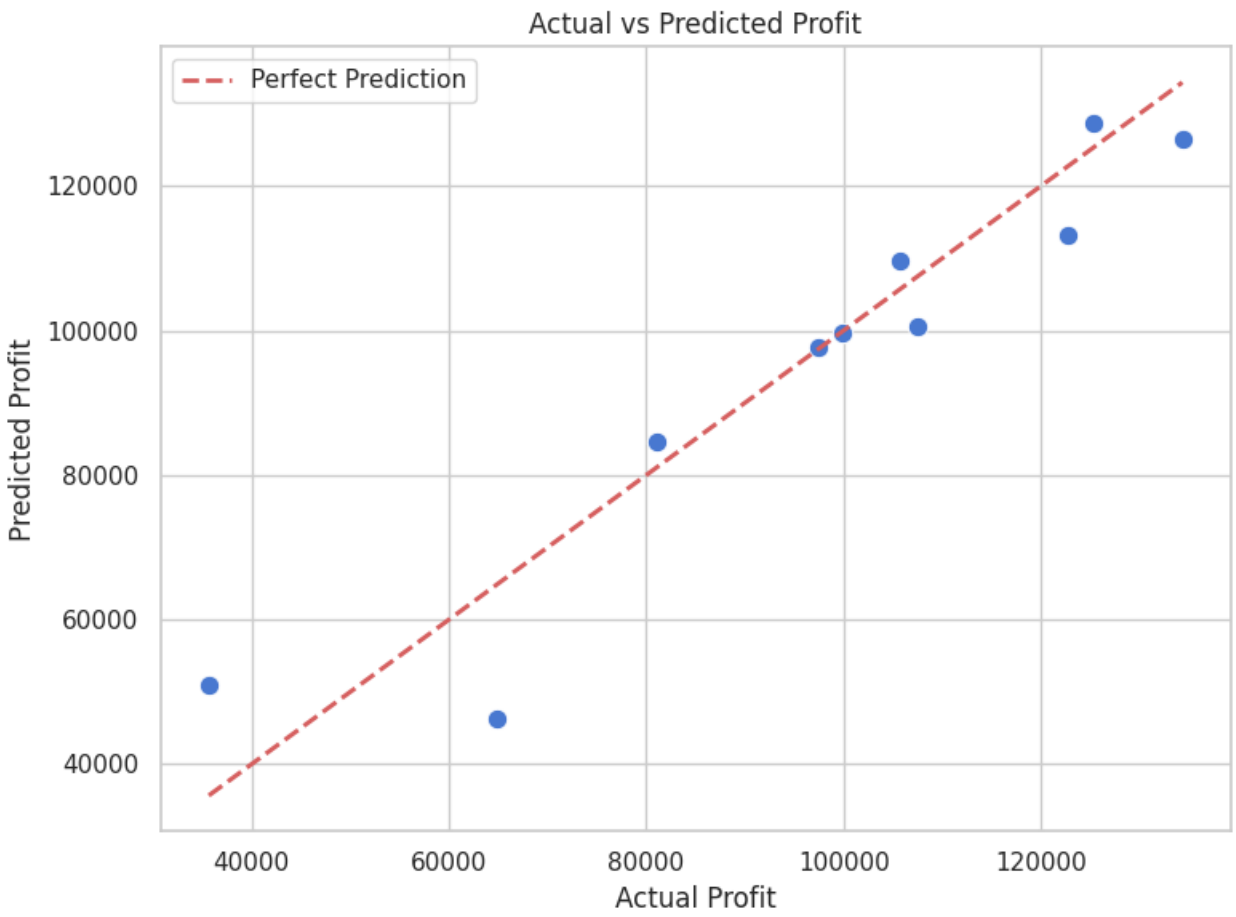
---

20 · Actual vs Predicted Plot

Points closer to the red dashed line indicate better predictions.

> ***AI Note:*** *This cell generates a scatter plot to visualize the relationship between actual and predicted profits, providing insights into the model's performance.*

```
plt.figure(figsize=(8, 6))
sns.scatterplot(x=y_test, y=y_pred, s=80)
plt.plot(
    [y_test.min(), y_test.max()],
    [y_test.min(), y_test.max()],
    '--r', linewidth=2, label='Perfect Prediction'
)
plt.xlabel('Actual Profit')
plt.ylabel('Predicted Profit')
plt.title('Actual vs Predicted Profit')
plt.legend()
plt.tight_layout()
plt.show()
```



> ***AI Note:*** *Provides context for residual analysis, setting the stage for evaluating the fit of a model.*

---

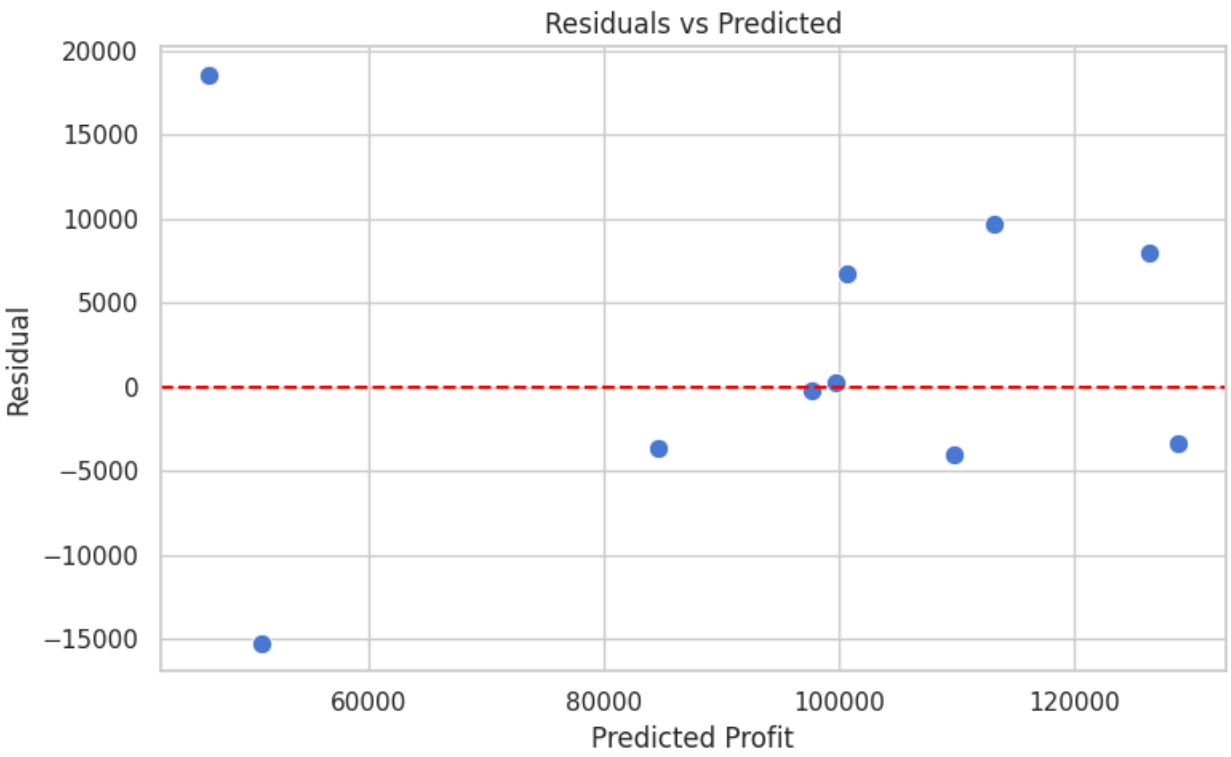
21 · Residual Analysis

Residuals should be randomly scattered around zero for a well-fitted model.

> ***AI Note:*** *This cell generates a scatter plot to visualize the residuals (actual - predicted) against the predicted values, providing insights into the model's fit and identifying potential issues.*

```
residuals = y_test - y_pred

plt.figure(figsize=(8, 5))
sns.scatterplot(x=y_pred, y=residuals, s=80)
plt.axhline(0, color='red', linestyle='--')
plt.xlabel('Predicted Profit')
plt.ylabel('Residual')
plt.title('Residuals vs Predicted')
plt.tight_layout()
plt.show()
```



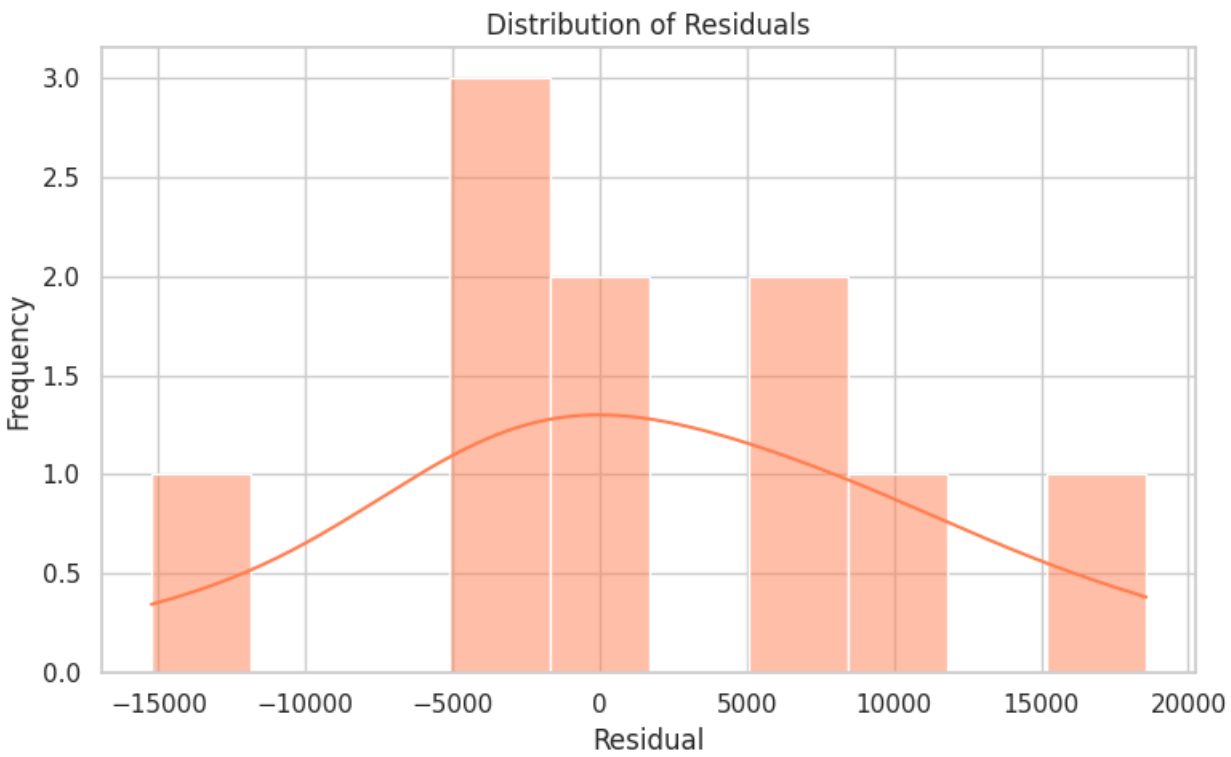
> ***AI Note:*** *Provides context for the residual distribution analysis.*

21.1 — Residual Distribution

> ***AI Note:*** *This cell generates a histogram to visualize the distribution of residuals, providing insights into the model's fit and identifying potential issues.*

```
plt.figure(figsize=(8, 5))
sns.histplot(residuals, kde=True, bins=10, color='coral')
plt.title('Distribution of Residuals')
plt.xlabel('Residual')
plt.ylabel('Frequency')
```

```
plt.tight_layout()
plt.show()
```



---

22 · Feature Importance

Model coefficients indicate how much each feature influences profit (holding others constant).

> ***AI Note:*** *This cell generates a DataFrame to display the coefficients of the linear regression model, sorted in descending order by their magnitude, providing insights into the relative importance of each feature in the model.*

```
coef_df = pd.DataFrame({
    'Feature': X.columns,
    'Coefficient': model.coef_
```

```
}).sort_values('Coefficient', ascending=False)

coef_df
```

Result:

```
            Feature  Coefficient
3   State_Istanbul   938.793006
4   State_Pakistan     6.987760
0        R&D Spend     0.805630
2  Marketing Spend     0.029855
1   Administration    -0.068788
```

> ***AI Note:*** *Provides a heading for the coefficient bar chart section.*

22.1 — Coefficient Bar Chart

> ***AI Note:*** *This cell generates a bar chart to visualize the feature importance of a linear regression model, showcasing the relative importance of each feature in the model.*

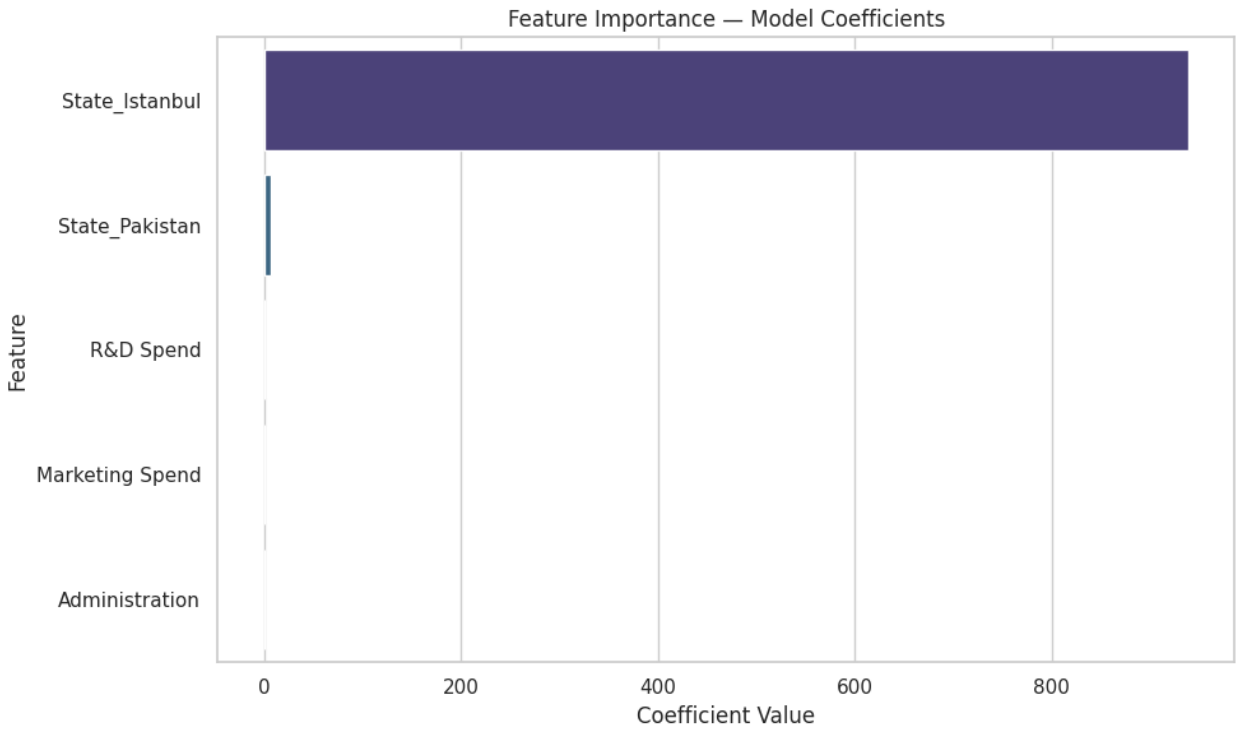
```
plt.figure(figsize=(10, 6))
sns.barplot(x='Coefficient', y='Feature', data=coef_df, palette='viridis')
plt.title('Feature Importance — Model Coefficients')
plt.xlabel('Coefficient Value')
plt.ylabel('Feature')
plt.tight_layout()
plt.show()
```

Output:

```
/tmp/ipykernel_70051/1288200969.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v

  sns.barplot(x='Coefficient', y='Feature', data=coef_df, palette='viridis')
```



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23 · Conclusion

Insight	Detail
Model Performance	The R^2 score quantifies how well our features explain Profit variance. MAE / RMSE give the average prediction error in dollar terms.
Key Driver	R&D Spend shows the strongest positive correlation and largest coefficient — it is the most important predictor of Profit.
Administration	

Insight	Detail
	Has a weaker (and potentially negative) relationship with Profit, suggesting over-spending on administration may not help.
Marketing Spend	Moderate positive influence; useful but secondary to R&D.
State	Minimal impact on Profit after encoding — geographic location is not a strong predictor.

Recommendation: A venture capital firm should prioritize startups that invest heavily in R&D, followed by Marketing — while keeping Administration costs lean.